

Front Matter

- **Title Page**
- **Abstract**
 - *Content:* Purpose of the project, MuJoCo/PPO methodology, and quantitative efficiency outcomes (e.g., target tracking accuracy and Cost of Transport percentage reduction).
- **Acknowledgements & Declarations**
- **Lists of Figures & Tables**

Chapter 1: Introduction

- **1.1 Background:** Legged robotics paradigm shift from classical control (such as Model Predictive Control) to end-to-end Deep Reinforcement Learning.
- **1.2 Problem Statement:** Standard reward-shaping techniques generate high-torque, structurally violent, and energy-inefficient gaits.
- **1.3 Project Scope & Deliverables:** Definition of targeted velocity tracking with energy constraint budgets on the Unitree A1 platform.
- **1.4 Research Contribution:** Exploring how multi-scale reward formulations and constrained optimizations prevent policy degradation under environmental noise.

Chapter 2: Theoretical Foundations & Literature Review

- **2.1 Markov Decision Processes (MDPs):** Mathematical description of continuous state spaces, action spaces, transition probabilities, and discounted returns.
- **2.2 Proximal Policy Optimization (PPO):** The clipped surrogate objective function:

$$L^{CLIP}(\theta) = \mathbb{E}_t \left[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t) \right]$$

Explain why clipping limits the policy update step size to maintain training stability.

- **2.3 Mechanics of Legged Locomotion:** Classical gait definitions (such as the trot, walk, and pace), ground reaction forces, joint torque physics, and the mathematical definition of **Cost of Transport (CoT)**:

$$\text{CoT} = \frac{P}{m \cdot g \cdot v}$$

where P is input power, m is mass, and v is base velocity.

- **2.4 Prior Work in Energy-Efficient RL:** Summary of historical approaches from top robotics laboratories (e.g., MIT Cheetah, ETH Zurich ANYmal) regarding minimizing control effort in legged systems.

Chapter 3: System Architecture & Simulation Interface

- **3.1 The MuJoCo Simulation Environment:** Technical explanation of MuJoCo's convex contact solver. Define the simulation timestep ($\Delta t_{sim} = 0.002$ s) and control frequency ($f_{control} = 50$ Hz) choices.
- **3.2 Kinematic Profile of the Unitree A1:** Physical parameters of the model (including mass of the trunk, link lengths, joint velocity constraints, and motor torque limits).
- **3.3 Observation Space Specification:** Mathematical description of the 30-dimensional normalized sensory array fed to the neural network:
 - Base orientation (quaternions) to avoid Euler gimbal lock.
 - Base angular and linear velocities.
 - Joint positions (q) and velocities ($\dot{q}$).
 - Previous joint action commands.
- **3.4 Action Space & Actuator Control:** Action vector mapping $[-1, 1]$ from the policy to target positions for the low-level Joint Proportional-Derivative (PD) controllers. Detailed layout of K_p and K_d gains.

Chapter 4: Reward Engineering & Learning Mechanics

- **4.1 Phase I: Targeted Velocity Tracking:** Detailed explanation of the multi-scale Gaussian reward function:
$$R_{velocity} = w_{broad} \cdot e^{-\alpha_{broad} (v - v_{ref})^2} + w_{tight} \cdot e^{-\alpha_{tight} (v - v_{ref})^2}$$
 - **The Math Rationale:** Why a broad Gaussian provides a continuous exploratory gradient for random actions, while a tight Gaussian enforces high-precision tracking at steady state.
- **4.2 Pitfalls of Reward Plateaus (The Flat Tolerance Zone):** Theoretical proof of why using step functions or "if-else" tolerance bands causes vanishing gradients, prompting the agent to settle on sloppy upper bounds.
- **4.3 Phase II: Energy Optimization Formulations:**
 - **Control Effort Penalty:** Minimizing joint torque squared ($\sum \tau_i^2$).
 - **Mechanical Power Penalty:** Tracking instantaneous work ($\sum |\tau_i \cdot \dot{\theta}_i|$).
- **4.4 Optimization Scaling & Avoidance of Policy Devolution:** The normalization and weighting strategies (w) designed to prevent the agent from adopting a "standing still" local optimum (the lazy agent trap).

Chapter 5: Experimental Evaluation & Results

- **5.1 Training Convergence Profiles:** High-resolution charts tracking reward accumulation, policy loss, and value loss over 5 million steps.
- **5.2 Gait Performance Benchmarking:** Side-by-side comparative tables of the Baseline Policy (Phase I) versus the Energy-Optimized Policy (Phase II).
 - Metrics to plot/table: Mean velocity tracking error, foot slippage, and torso orientation variance (gait smoothness).
- **5.3 Energetic Efficiency Analysis:** Quantitative plots demonstrating the reduction in average joint torque and the resultant decline in Cost of Transport (CoT).
- **5.4 Phase III: Environmental Robustness Evaluations:**
 - Gait adaptation profiles when ground friction drops dynamically from $\mu = 1.0$ to $\mu = 0.2$ (simulating ice).
 - Force perturbation tracking: Plots showing the recovery trajectories of the center of mass when subjected to instantaneous sideways impulse force vectors (pushes of 10N, 20N, and 30N).

Chapter 6: The Sim-to-Real Chasm (Discussion)

- **6.1 Unmodeled Simulator Dynamics:** Identifying what the MuJoCo model leaves out (such as motor latency, gear backlash, friction hysteresis, and battery voltage sag).
- **6.2 System Identification Methodologies:** Explaining how sensor calibration and parameter matching would be executed on a physical Unitree A1 to minimize this gap.
- **6.3 Domain Randomization Framework:** Outline of the randomized physics variables (mass variations, ground friction, and sensor noise) required to make the simulation policy deployable on physical hardware.

Chapter 7: Conclusions & Future Work

- **7.1 Project Summary:** Review of the implementation milestones and the validity of our research hypothesis.
- **7.2 Key Contributions:** Demonstrating how multi-scale Gaussian rewards, paired with structured energy costs, create efficient, smooth, and robust robotic controllers without triggering policy degradation.
- **7.3 Future Recommendations:** Extending this framework to uneven terrains (such as stairs or slopes) and transitioning to 3D visual perception inputs.